

LOGICAL REASONING TEST STUDY GUIDE

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Test Purpose

This is a test designed to evaluate your logical reasoning ability. These tests will challenge you to solve abstract problems using a wide variety of skills, thought processes, and rational exercises. Unlike some other tests, each question here DOES have right and wrong answers and each answer is accessible through the correct form of logical reasoning.

Test Format

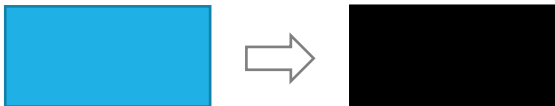
The question types vary, but many of the categories include figures or shapes in patterns that must be identified and analyzed in order to get to the right answer. For other types of questions, there will be numbers and some simple math required to identify patterns and correctly answer the questions.

Inductive and Deductive Reasoning

This test will ask you to use two basic types of logic in solving the questions; Inductive and Deductive Reasoning. While these may sound complicated, they are just ways of thinking through problems to get the right answers. You already do these processes automatically, but it is important to know what is required of you for the test.

Inductive Reasoning

This type of logic is based first off of making observations. Once you make an observation, you know that this observation is true. From there, you can induce logic towards a broader conclusion that is based off of the true observations you have made. Think of it as “bottom-up” logic: You are going from the specific, small details, and working your way up toward broader conclusions. Here is a simple example:



- You have observed that the color of the shape has changed.
- You have observed that the shape, size, angle or number of shapes has not changed.
- Therefore, color is the changing variable.
 - Being able to identify the changing variable is key to these questions. If you can find the changing variable, you know what to pay attention to and you can solve these puzzles much easier.

CHANGING VARIABLES in Inductive Reasoning:

Above, you have seen color change as an example of a changing variable. There can be many other variables that change as well. Keep the following list in mind when you are analyzing figures and shapes on this test:

- Color Change (Did the shapes change color?).
- Number Change (Did the number of shapes change between the two pictures?).
- Angle rotation (Did the shapes turn, move, or shift angles?).
- Mirroring (Is the shape flipped, as if you were looking in a mirror? Remember, shapes can flip in two directions- vertically and horizontally.).
- Movement (Where is the shape in the picture? Did it move?).
- Size (Did the shape get bigger? Smaller?).
- Folding/Symmetry (Did the shape fold in some way while still maintaining its integrity?).

Deductive Reasoning

If Inductive Reasoning is "bottom-up" logic, Deductive Reasoning is "top-down" logic. Deductive logic starts with a "premise", which is usually a fairly broad statement. Once you have accepted the premise as true, it is your task to deduce toward a more specific, detailed conclusion. The basis of truth in this case is having a valid premise to begin with. Here is the classic traditional example of deductive reasoning (which is a basic form of Syllogism).

- If Socrates is a man
- And all men are mortal
- Then Socrates is mortal

These types of questions can be confusing, but all must follow the same simple formula. Let's look at these three statements again, this time with each statement labeled.

- If Socrates is a man (A)
- And all men are mortal (B)
- Then Socrates is mortal (C)

In order for any of this to be true, A must equal B ($A=B$). Sometimes, logic questions can be solved by proving that A does NOT equal B, therefore that logic is invalid (think of this when trying to eliminate incorrect answers).

The basic formula for syllogism questions is this:

If $A=B$

And $B=C$

Then $A=C$.

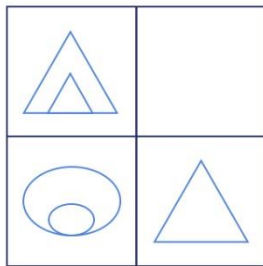
Questions will get much more complicated and involve many more variables than that, but the principles of deductive reasoning still apply. You start with "truths" on broad premises, and you can deduce logic towards an answer using the formula above.

Question Types

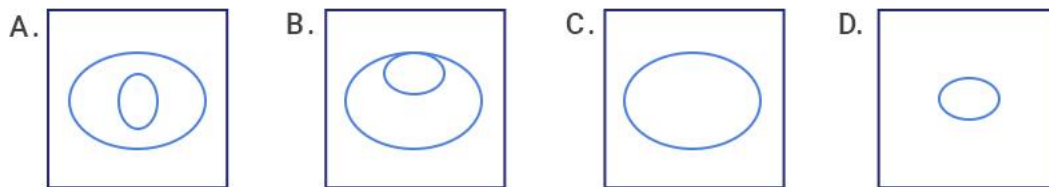
Figural Analogies

Figural Analogy questions are testing your ability to identify relationships between different figures, and then to identify patterns within those relationships. These questions can be incredibly simple, but also can be incredibly difficult.

Example:



Which of the following options below belong in the blank square above?



The correct answer to this question is C. The relationship and pattern here is that there are smaller, matching shapes inside the larger shape in the first column, and then there is a corresponding shape that does not have the inner shape in the second column. The triangle shows us this pattern, so the logical conclusion is that C, the circle with no inner circle, is the right answer.

Tips:

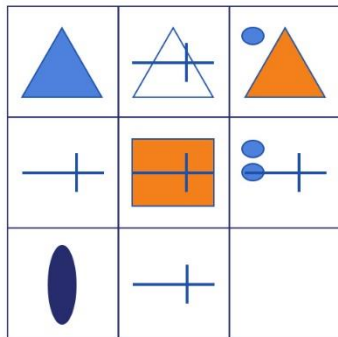
- Take your time and look closely. Make sure you study each element of the figures and patterns. There are often hidden things in these figures that you might not catch on first glance.
- The answers are right in front of you! This is true in more ways than one.
 - If there are 4 answer choices given, use those to help you identify the pattern.
 - Multiple choice answers are part of the question- READ THEM and use them to help you solve.

- All of these questions are solvable by logic. If you know the pattern, these are easy questions! Only guess if you are truly stumped. Most of the time, you will be able to look at the question long enough and crack it.

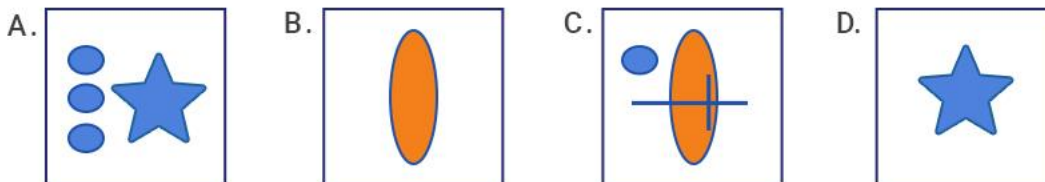
Figural Matrices

Figural matrices are a group of shapes in a rectangular matrix that are related to each other, and ultimately have a solution or pattern based on the development of the figures through the matrix. These questions are similar to the figural analogy questions. However, these are often more complicated with more going on in each figure. This means it is very easy to get distracted, or have trouble finding the pattern through the many shapes within the figure. As with all of these questions, the key to success is finding the patterns.

Example:



Which of the following options belong in the blank square above?



This is a difficult question because there are a lot of distractions. However, the only discernible pattern here is the increasing dots along the right column. In the first row, there is a shape with one dot. In the second row, the shape is accompanied by two dots. So the third row must have a shape accompanied by three dots, which corresponds with answer A.

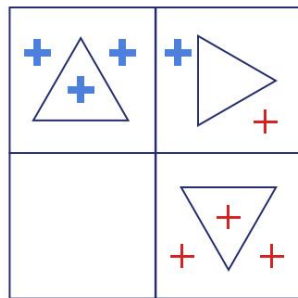
Tips:

- Focus on two figures at a time. Taking in the whole matrix can be overwhelming, so focus on comparing just two next to each other. This makes it much easier to find differences and identify potential patterns.
- The answers are right in front of you! This is true in more ways than one.
 - If there are 4 answer choices given, use those to help you identify the pattern.
 - Multiple choice answers are part of the question- READ THEM and use them to help you solve.
 - All of these questions are solvable by logic. If you know the pattern, these are easy questions! Only guess if you are truly stumped. Most of the time, you will be able to look at the question long enough and crack it.

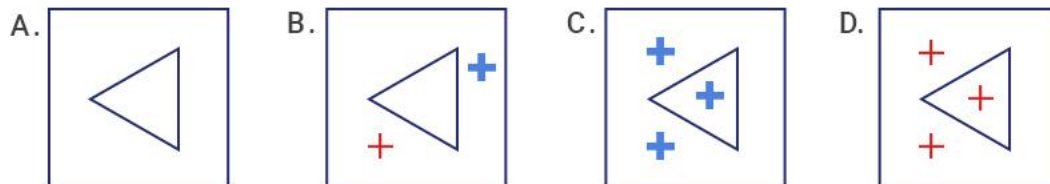
Figural Next In-Series

Figural next in series questions are asking for a very specific type of analysis and only require one skill. You must identify a series- a pattern that is predictable and has a logical “next” in the sequence. This requires the evaluation of different shapes and relationships between shapes within a square and shapes between squares.

Example:



Which of the following options belong in the blank square above?



The correct answer here is B, because it satisfies all the discernible patterns. First, you have a rotating triangle. Notice that all of the correct answers have the triangle pointed in the correct direction. Next, you must look at the crosses in relationship to the triangle. In the top row, there are three blue crosses, followed by a triangle with one blue and one red. In the bottom row, there are three red crosses, which indicates you should be looking for two crosses, one red and one blue. B has this and is the correct answer.

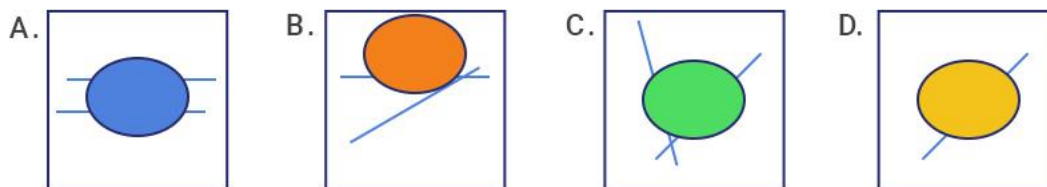
Tips:

- Remember that this is a series! The patterns and relationships exist, and you know that it is leading you towards a resolution. Since you know the series ends in the blank square or space, you have linear movement toward the end of that series. This could help you sort through the answers and find the right one
- Having trouble finding the pattern? Plug in each of the multiple-choice answers and see by trial and error if you can find the right solution, or at least identify the pattern in the other squares.

Figural Odd One Out

Figural Odd One Out questions ask you to identify a pattern between figures and find the one that clearly does not fit that pattern. This is another type of skill being asked for here, as you not only have to identify the pattern that fits for three, but also find one that does not fit the pattern.

Example:



Which of the following options does not fit with the other three?

The answer to this odd one out is fairly simple. D has a circle with only one line bisecting it, while all of the others have 2 lines going through the circle.

Tips:

- The easiest thing to do for these questions is to find the commonality between three, and then figure out the difference for the fourth. Since three will all have some element in common, that will be the easiest thing to spot when looking for a pattern.
- Don't get distracted! Often, these questions try to trick you, when there is a simple and obvious difference between one of the figures. Try to focus on similarities and differences and not the distractors.

Figural A/B Sets

Figural A/B Sets can appear tricky at first, but are actually quite simple. You have two choices here, subtraction or combination. In both cases, the key first step is to identify numbers, letters or names that appear in both sets. If you are subtracting, take the set being subtracted out of the primary set. For example, if the question asks A-B, you would remove all the names that appear in B and A, and remove them from A. Thus, the right solution is only what remains in set A. For a combination problem, simply see what values are repeated, and only include those in the correct answer once. So AB would be a merger of all values, without repeating values.

Example:

<u>Set A</u>	<u>Set B</u>
1, 3, 5, 7, 9	3, 4, 5, 6, 7

Solve: A-B

The answer here is 1, 9. You are subtracting B from A. So you must eliminate all repeated values that occur in both A and B from A. Thus, 3, 5 and 7 are eliminated, leaving only 1 and 9.

Tips:

- Practice and get used to the questions. These can seem confusing at first, but once you figure out the trick, these are incredibly easy.
- Be careful. Make sure you are subtracting or combining according to what the question asks, and make sure you match up the figures correctly when looking for repeat values. These questions lend themselves to simple mistakes.

- There should be no guesswork here. As long as you are careful and are doing the correct operation (which also should come down to care and attention) you should get these questions correct.

Syllogism

Syllogism is classical logic. If $A=B$, and $B=C$, then $A=C$ is the fundamental principle you need to remember here. These questions are all solvable using syllogism like the formula listed above. They will be infinitely more complicated than that simple formula, but the principle remains the same. In most cases, these questions will contain nonsense, made up words as names for things so that you have to focus on the logic of a statement and not allow outside knowledge of things or events aid you in answering the questions.

Example:

A BROG is similar to a GRANGY but longer and thinner. A FUVET is like a DRANGO but is shorter and thicker. A DRANGO is like a GRANGY but is shorter and thinner.

Statement: A BROG is like a DRANGO only much longer and thinner.

- True
- False
- Cannot Say

The answer here is cannot say. The statement clearly states that a BROG is similar to a GRANGY only longer and thinner, and that a DRANGO is shorter and thinner than a GRANGY. Thus, the statement about BROGS being longer than DRANGOS is true, however we cannot say if BROGS are thinner than DRANGOS. Both BROGS and DRANGOS are thinner than GRANGYS, however there is nothing in the statement that says whether BROGS or DRANGOS are thinner.

Tips:

- Break it down one statement at a time.
- Write down things to help you organize and keep it straight. It may be too much to have in your head, but by writing things down you can keep it straight and arrive at the correct answer.
- Do not stray from your syllogism principles. Do not use outside knowledge or perception to answer the questions, only what is presented.
- REMEMBER: If $A=B$, and $B=C$, then $A=C$.

Seating Arrangements/ Logical Organization

These questions require a similar type of thinking as syllogism, but look very different in how the questions are asked. These questions will ask you to logically analyze a situation given certain parameters and find the best, most logical solution that meets all the requirements.

Example:

First period history class has seven students, and the teacher must make a seating arrangement based on the following requirements:

- Students 1, 2, & 3 are girls, while student's 4, 5, 6, & 7 are boys. Your seating arrangement must alternate boy/girl.
- Students 2 and 4 just broke up, and cannot sit next to each other.
- Students 6 and 7 are best friends and can't stop talking, so they need to be placed as far away from each other as possible.
- Students 3 and 5 are both excellent students and can help focus the weaker students.
- Students 2 and 7 have been close friends since elementary school and have requested to sit together.

Which seating arrangement should you choose?

- A: 4,2,7,3,6,1,5
- B: 7,2,5,1,4,3,6
- C: 2,5,1,7,3,4,6
- D: 6,1,7,2,5,3,4

B is the correct response here, as it satisfies all the requirements given in the question.

Tips:

- If given multiple choice answers, evaluate each answer and look for places where it violates one of the parameters set out. The right answer will not violate any of the rules, so you can eliminate answers easily that you find a problem with. This will almost assuredly cut down the possible answers to either three or two choices, and perhaps even leave you with the one correct answer.
 - Process of elimination is always your friend!
 - With multiple choice answers given here, you don't have to re-invent the wheel!
- Always double check to make sure your chosen answer does not have a problem. There are so many moving parts to these questions and rules to follow; it is easy to overlook something. Double-check to make sure you don't lose a point on a silly mistake!
- Write things out as you read through the information. Once again, there are a lot of components here, and writing things down will help you sort through it all.
- It is ok to use trial and error. Try to make a seating chart by following the rules, and then you can go back and tweak it if you notice problems. This is a great strategy if you like to work on things actively and hands-on.

Inductive Reasoning

Inductive reasoning is when one begins with observations and using those observations and logical generalizations, one can come to conclusions based off of verifiable observation. These types of questions can be confusing because it is hard to "prove", but each question has an answer that is best and "most logical" based on the information you have been given.

Example:

You are searching for a new apartment and have the following criteria in what you want in your new home.

- 2 bedrooms (highest priority)
- 1 bathroom
- Modern appliances in kitchen
- Wood floors
- Pet friendly

You have visited the four apartments below:

APT #1: 1 bed, 1 bath, modern appliances, including washer and dryer in unit, wood floors and pet friendly.

APT #2: 2 bed, 1 bath, wood floors, pet friendly, washer and dryer in unit.

APT #3: 2 bed, 2 bath, wood floors, modern appliances, no washer and dryer in unit.

APT #4: 2 bed, 2 bath, wood floors, modern appliances, no washer and dryer in unit, pet friendly.

Of the 4 choices listed above, which apartment should you choose?

Explanation: Apt #4 is the correct answer because it meets all the criteria listed in the prompt. The only negative is not having a washer/dryer in unit, but that is not listed on the required criteria. Apartments 1, 2, and 3 all have at least one criteria they do not meet, leaving apt 4 as the correct answer.

Tips:

- FOCUS on the FACTS
 - The test provides you with enough factual information for you to get to the best answer. In order to make the right logical conclusions, you must read and understand all of the facts to properly contextualize your logic.
- Trust your instincts and common sense
 - Do not overthink these questions; the answers are usually fairly straightforward and easy to understand through a logical series of thoughts.
- Check your answers back against the facts
 - Does your answer choice align with the facts given? Double-check to make sure you are following all of the rules!
- Use written aids to help you
 - These questions can often have a lot of information and can be hard to take in. Do not hesitate to write notes or make some kind of graphic organizer to keep all of the information straight.

Deductive Reasoning

Deductive reasoning, opposite of inductive reasoning, is when one begins with a broad theory and using logical reasoning, one can deduce a fact. These types of questions can be hard because they can seem vague and difficult at first, and you may think “how am I supposed to know” or even “that is an opinion, not a fact.” Remember, each question DOES have a correct answer. This is yet another case where you need to choose the “best” answer using logical reasoning.

Example:

After the last day of the season, there was a three way tie at the top of the basketball league standings. The Hawks, Wildcats, and Tigers all finished with the same record 8-2. Here are the league tiebreaker rules:

- In the event of a tie in terms of final record- head to head results will determine who finishes higher in the standings.
- In the event of a three way tie, the record against the other top teams will be considered the tiebreaker.
- If the teams are still tied, point differential in the games between the other two teams will be used to break the tie.

Many coaches in the league are very unhappy with this tiebreaker rule because they feel it encourages poor sportsmanship. Which of the following statements best encompasses these coaches' arguments?

- A. The rules are unfair because they favor higher scoring teams.
- B. The rules encourage teams to run up the scores on other teams.
- C. The tiebreaker should take into account record against the fourth place teams rather than focus on differential.
- D. The tiebreaker should not exist, and teams should share the championship.

Explanation: B is the correct answer. That is the argument that is directly tied to sportsmanship, and since the tiebreaker values point differential, it is a valid complaint because it does encourage running up the score.

Tips:

- Trust your instincts and common sense.

- Do not overthink these questions; the answers are usually fairly straightforward and easy to understand through a logical series of thoughts.
- You need to read and understand the facts in order to think logically to answer the question. Read the facts, but the trick here is proper logic, so reading and understanding the answer choices is really the key.
- Use written aids to help you.
 - These questions can be confusing and hard to keep straight in your head. It would make sense to make logic maps in order to track your thinking and make sure your logic is sound to arrive at the right answer.
- REMEMBER- your task is to DEDUCE.
 - Take what you are given and deduce the correct answer. While this is not easy, remember what the question is asking you to do and don't get bogged down in worrying about things unrelated to the question. Focus on using deductive reasoning to arrive at a logical conclusion.